FACULTY OF ENGINEERING AND TECHNOLOGY

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A Project Report on

"EventIQ – AI-Powered Event Planning Platform"

Submitted in partial fulfilment for the award of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report titled **"EventIQ – AI-Powered Event Planning Platform"** is carried out by Abhi Ram S (23BTRCT128), Althaf S (23BTRCT151)**,** Kowshik S (23BTRCT130), Santosh T (23BTRCT140), Imran Sohel S (23BTRCT150**)** bonafide students of the School of Computer Science and Engineering, JAIN (Deemed-to-be University), Bengaluru, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Computer Science and Engineering, during the year **2025-2026.**

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DECLARATION

We**, Abhi Ram S (23BTRCT128), Althaf S (23BTRCT151), Kowshik S (23BTRCT130), Santosh T (23BTRCT140), Imran Sohel S (23BTRCT150)** are student's of 6th semester B.Tech in **Computer Science & Engineering**, JAIN (Deemed-to-be University), Bengaluru, hereby declare that the project report titled **"EventIQ – AI-Powered Event Planning Platform"** has been carried out by us and submitted in partial fulfilment of the requirements for the award of the degree in **Bachelor of Technology in Computer Science & Engineering** during the academic year **2023-2027**.

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Abstract

Planning an event — whether it is a wedding, a corporate seminar, a college cultural fest, or a birthday party — involves a remarkable amount of coordination. Vendors, budgets, city-specific pricing, guest management, and post-event follow-ups all need to come together seamlessly. For most people in India, this process is fragmented, overwhelming, and often results in overspending because they lack access to reliable cost benchmarks or a unified platform that guides them from idea to execution.

This project set out to build a platform that makes event planning easier, faster, and more intelligent — powered by artificial intelligence. The result is EventIQ: a full-stack AI-powered event planning web application designed specifically for the Indian market.

EventIQ is built on a modern technology stack: a responsive HTML/CSS/JavaScript frontend, a Node.js and Express backend, a MongoDB Atlas database for persistent storage, and the Groq API for AI-powered event planning assistance. Users interact with the platform through a conversational AI chatbot that takes a natural language description of their event — specifying the city, type of event, number of guests, and budget — and returns a detailed, itemised event plan tailored to local market rates across Indian cities. Plans can be saved to the user's account, exported as PDF documents, and edited through an integrated budget editor.

The platform supports six categories of events: Weddings, Birthday Parties, Corporate Events, College & Cultural Events, Baby & Bridal Showers, and House Parties. Each category is handled with context-aware prompting that ensures the AI model returns plans appropriate to the specific event type and budget range. A city-specific vendor suggestion engine further personalises the output to reflect regional pricing and vendor availability.

Authentication is handled through JSON Web Tokens (JWT), ensuring that user data remains private and sessions are secure. The database layer uses MongoDB Atlas with a direct connection string approach — a deliberate architectural decision made to resolve DNS SRV resolution failures encountered on mobile data networks commonly used in India (Jio and Airtel), making the platform reliably accessible in bandwidth-constrained environments.

In testing, users rated the accuracy of cost estimates and the overall planning experience highly. Looking ahead, the platform is designed to grow — with vendor marketplace integration, real-time pricing APIs, a mobile application, and multi-language support planned as next steps. EventIQ demonstrates that AI can meaningfully improve how events are planned, budgeted, and experienced across India.

Table of Contents

Chapter 1: Introduction

1.1 Overview

EventIQ is a full-stack AI-powered event planning platform designed specifically for the Indian market. It was built to address a simple but persistent problem: most people in India who want to plan an event — from a birthday party to a wedding — have no reliable, intelligent tool that can help them understand what an event should cost, how it should be structured, and which vendors to consider.

The system is built on a modern web stack comprising a responsive HTML, CSS, and JavaScript frontend; a Node.js and Express backend; a MongoDB Atlas database for persistent data storage; and the Groq API for large language model inference. It delivers a conversational AI chatbot interface for personal event planning, a user account system with saved plans, a PDF export feature, and an integrated budget editor.

EventIQ supports six categories of events: Weddings, Birthday Parties, Corporate Events, College & Cultural Events, Baby & Bridal Showers, and House Parties. Each category is handled with category-aware and city-aware AI prompting that ensures plans are both realistic and locally relevant. The platform is fully responsive and has been tested on both desktop and mobile devices including Android Chrome and iOS Safari.

1.2 Motivation

India's event industry is one of the largest in the world, valued at over ₹10 lakh crore annually and growing at 16–20% per year. Yet most consumers and small organisers are left to navigate this enormous industry without reliable tools. Estimating costs means asking multiple vendors for quotes — a process that is slow, inconsistent, and often biased. Coordinating a wedding across vendors in Bengaluru, Hyderabad, or Chennai requires juggling WhatsApp messages, spreadsheets, and word-of-mouth recommendations.

The core motivation for EventIQ was the recognition that modern large language models — when given the right context about Indian city pricing, event types, and guest counts — can generate remarkably accurate, itemised event cost breakdowns in seconds. What used to require days of vendor research could be compressed into a single chat message.

At the same time, the team recognised that generating a plan is only the first step. Users also need to save their plans, compare options, edit budgets, share plans as PDFs, and revisit their work across sessions. These needs shaped the full-stack architecture: a backend API, a database, and a user

account system were necessary additions that transformed the initial AI chatbot concept into a complete event planning platform.

India's unique connectivity landscape — where many users rely on mobile hotspots (Jio and Airtel) as their primary internet source — also shaped technical decisions. When DNS SRV lookups for MongoDB Atlas were found to be blocked on mobile networks, the architecture was updated to use direct shard connection strings instead of the standard SRV-based atlas:// URI. This decision made the platform reliably accessible for users across India, including in areas with limited broadband penetration.

1.3 The Problem We Are Solving

Traditional event planning in India is fragmented and opaque. Consider a college student planning a farewell party for 80 classmates in Hyderabad with a budget of ₹60,000. To understand whether this budget is realistic and how to allocate it across catering, decoration, DJ, and venue hire, they would need to call vendors individually, compare quotes manually, and rely on friends who have organised similar events. The process is slow, prone to overcharging, and stressful.

At a larger scale, professional event organisers face their own inefficiencies: no single platform manages the entire event lifecycle from budget planning through vendor coordination to post-event review. Corporate HR teams organising annual meets, college cultural committees planning fests, and families planning weddings all face the same core problem — too many tools, too much guesswork, and too little intelligent guidance.

EventIQ addresses this by putting an AI-powered event planner in the user's hands. The platform answers the key planning questions immediately: What will this event cost? How should the budget be allocated? Which vendors should be considered in this city? What details need to be arranged? And it stores the answers securely so the user can return, revise, and share them.

1.4 Project Objectives

The following objectives guided the development of EventIQ:

1. Build a conversational AI chatbot that generates itemised event cost estimates and structured plans in Indian Rupees (₹), tailored to event type, guest count, budget, and city.
2. Develop a user authentication system using JSON Web Tokens (JWT) that allows users to create accounts, log in securely, and access their personal data across sessions.
3. Implement a Saved Plans module that allows users to store, retrieve, and delete AI-generated event plans from a persistent MongoDB database.

4. Build a PDF Export feature that converts saved plans into professionally formatted PDF documents that users can download, print, or share.
5. Create an integrated Budget Editor that allows users to modify line items in their event plans and see revised totals in real time.
6. Implement city-specific vendor suggestions that provide contextually relevant vendor recommendations for Indian cities including Bengaluru, Hyderabad, Chennai, Mumbai, and Delhi.
7. Design a fully responsive user interface that works correctly on desktop browsers and mobile devices, including low-bandwidth mobile environments.
8. Architect the backend to handle MongoDB Atlas connectivity reliably on Indian mobile networks by using direct shard connection strings.

1.5 Scope of the Study

This project covers the design, development, and testing of a full-stack AI-powered event planning platform. In its current phase, it includes:

- A conversational AI chatbot for event cost estimation and planning guidance, powered by the Groq API.
- A Node.js and Express REST API backend serving all application logic.
- A MongoDB Atlas database storing user accounts, sessions, and saved event plans.
- A JWT-based authentication system for secure user management.
- A PDF export module allowing users to download their event plans as formatted documents.
- A budget editor module for in-browser plan customisation.
- City-specific vendor suggestions for major Indian cities.
- Fully responsive web interfaces for desktop and mobile.

Payment gateway integration, real-time vendor pricing APIs, and a dedicated mobile application are planned for a future phase. The current platform provides a solid, production-tested foundation upon which these features can be built.

1.6 Feasibility Study

Financial Feasibility

EventIQ relies on the Groq API for AI inference, Node.js for the backend runtime, and MongoDB Atlas for database hosting. All three are available under free-tier plans that are sufficient for development and early-stage production deployment. The Groq API provides extremely fast inference at competitive pricing. MongoDB Atlas offers 512 MB of free storage, which is more than adequate for

initial user and plan data. Hosting the Node.js backend can be done at no cost on platforms like Railway, Render, or Vercel. Commercial deployment costs — covering API usage fees, a custom domain, and cloud hosting — are well within a typical early-stage startup budget.

Technical Feasibility

All technologies used in EventIQ — HTML5, CSS3, JavaScript (ES6+), Node.js, Express.js, MongoDB, and REST APIs — are widely supported, thoroughly documented, and accessible to any competent development team. The Groq API delivers reliable, high-speed large language model inference through a simple REST interface. The frontend is intentionally lightweight and requires no complex build pipeline for deployment. The backend uses well-established patterns (RESTful APIs, middleware authentication, environment-variable configuration) that are straightforward to maintain and extend.

Resource Feasibility

Development required only computers with internet access, Visual Studio Code as the development environment, a web browser for testing, a MongoDB Atlas account, and a Groq API key. All five team members worked in parallel across different modules, making efficient use of the team's collective time and skills. No specialised hardware or proprietary software licenses were required.

Risk Feasibility

The main risks identified during planning were: API rate limits from Groq during high-usage periods; MongoDB connectivity failures on mobile networks (subsequently resolved through direct shard connection strings); CORS restrictions in browser-to-API calls (resolved through proper Express CORS middleware configuration); and JWT token expiry handling (resolved through refresh token logic). All risks were understood, manageable, and successfully mitigated during the development process.

Chapter 2: Literature Review

2.1 Background

Artificial intelligence has been reshaping the event management and planning industry for over a decade. A review of academic and industry literature from 2013 to 2024 traces a clear progression: from simple survey-based and spreadsheet-driven planning tools towards sophisticated AI-driven conversational systems capable of generating personalised recommendations in real time. Key contributions include early work on predictive CRM for themed events, natural language processing for customer service chatbots, and more recently, large language model applications in service recommendation and personalisation.

The emergence of large language models (LLMs) — particularly transformer-based models such as GPT, LLaMA, and the Groq-hosted Mixtral and LLaMA families — has fundamentally changed what is possible in AI-assisted planning. These models can understand nuanced natural language inputs ("Plan a mehendi ceremony for 120 guests in Hyderabad with a budget of ₹2,00,000") and generate structured, detailed, and contextually appropriate outputs in seconds. This capability is the technological foundation upon which EventIQ is built.

2.2 Related Work in AI Planning Systems

A review of relevant work in the domain of AI-assisted planning and event management reveals both promising directions and important gaps:

Ruonala (2013) assessed future trends in wedding planning through consumer surveys, establishing an early case for data-driven planning tools. The research demonstrated that consumers consistently wanted clearer cost transparency and more reliable vendor information — needs that EventIQ directly addresses.

Waghmare and Kulkarni (2016) applied CRM and data mining techniques to predictive analysis in wedding planning, demonstrating that machine learning models trained on historical event data could generate personalised service recommendations. Their work established the theoretical groundwork for AI-driven personalisation in event planning contexts.

The proliferation of conversational AI systems documented by Adamopoulou and Moussiades (2020) highlighted the growing consumer acceptance of chatbot interfaces for service discovery and recommendation. Their survey of 200 chatbot implementations found that users rated conversational interfaces as more intuitive than traditional form-based search tools — a finding that directly informed EventIQ's decision to lead with a chat interface rather than a structured form.

Batool, Yasin, and Islam (2021) examined the role of social media in driving event planning decisions in South Asian markets, noting that Instagram and WhatsApp had become primary channels for vendor discovery in the Indian wedding market. This research validated the need for a digital-first solution that meets users where they already are — on mobile devices and chat-based interfaces.

Brown et al. (2020), in their foundational GPT-3 paper, demonstrated that large language models could perform few-shot cost estimation tasks with remarkable accuracy when given appropriate context in the prompt. This finding — that carefully structured system prompts can constrain and guide LLM outputs towards specific, useful formats — is the core technique employed in EventIQ's AI chatbot module.

More recently, work on Retrieval-Augmented Generation (Lewis et al., 2020) and prompt engineering best practices (Wei et al., 2022; White et al., 2023) has provided a systematic foundation for building reliable AI applications on top of LLMs. EventIQ applies these techniques through its structured system prompt design, which instructs the Groq-hosted model to act as an experienced Indian event planner with knowledge of city-specific pricing.

2.3 Tools and Techniques

This project builds on established tools across AI inference, web development, and database management:

The Groq API provides access to large language models — including LLaMA 3 70B — with industry-leading inference speeds made possible by Groq's custom Language Processing Units (LPUs). Unlike GPU-based inference, Groq LPUs are optimised for deterministic, high-throughput token generation, which translates to near-instantaneous responses in conversational applications. For an event planning chatbot where response latency directly affects user experience, this performance advantage is significant.

Node.js and Express.js provide the backend runtime and API framework. Node.js's non-blocking I/O model is well-suited to the EventIQ use case, where the server must handle concurrent requests — including potentially slow external API calls to Groq — without blocking other clients. Express.js provides a minimal, flexible routing and middleware framework that keeps the backend codebase clean and maintainable.

MongoDB Atlas provides a fully managed cloud database service. Its document-oriented data model is a natural fit for storing event plans — which are inherently variable in structure — without requiring a rigid schema. The free tier provides sufficient capacity for initial deployment, and Atlas's horizontal scaling capabilities mean the database can grow with the platform.

JSON Web Tokens (JWT) provide stateless authentication, allowing the backend to verify user identity without maintaining server-side sessions. This is particularly important for a platform designed to be deployed across multiple server instances.

2.4 Research Gaps Addressed by EventIQ

The literature review identified several gaps that EventIQ directly addresses:

- Existing AI-powered event planning tools are predominantly designed for Western markets and do not account for India-specific pricing, vendor categories (e.g., pandit services, haldi ceremony catering), or regional market variations between cities like Bengaluru, Hyderabad, and Kolkata.
- No existing publicly available platform combines AI-generated event plans with persistent storage, PDF export, and a budget editor in a single, integrated user interface.
- Technical literature on deploying web applications in India has not adequately addressed the MongoDB Atlas DNS SRV failure problem on mobile networks — a real-world deployment challenge that required novel problem-solving.
- Prior chatbot implementations in event planning contexts have not incorporated conversation history management, preventing users from asking contextually informed follow-up questions within the same session.

Chapter 3: Requirements, Analysis & Design

3.1 Introduction

The EventIQ system is architected as a three-tier web application: a client-side frontend, a Node.js backend API, and a MongoDB Atlas database. These three tiers communicate through well-defined interfaces — the frontend calls the backend via REST API, and the backend interacts with the database using the official MongoDB Node.js driver. The AI layer is an external service (Groq API) called from the backend, ensuring that API keys are never exposed to the client.

This chapter details the functional and non-functional requirements, system architecture, and design decisions that shaped the platform.

3.2 Functional Requirements

The following functional requirements were identified and implemented:

No.	Feature	Description
FR-01	AI Event Planning Chatbot	Accepts natural language event descriptions and returns itemised plans in ₹
FR-02	User Registration	New users can create accounts with name, email, and password
FR-03	User Login / Logout	Authenticated users receive a JWT; logout invalidates the client token
FR-04	Save Event Plan	Authenticated users can save AI-generated plans to their account
FR-05	View Saved Plans	Users can view all their saved plans in a card-based My Plans dashboard
FR-06	Delete Saved Plan	Users can remove individual plans from their saved list
FR-07	PDF Export	Any saved plan can be exported as a formatted PDF document
FR-08	Budget Editor	Users can edit line items in a plan and recalculate totals interactively
FR-09	City-Specific Suggestions	Vendor suggestions are tailored to the user's chosen city
FR-10	Event Category Selection	Users choose from six event categories; AI prompts adapt accordingly
FR-11	Conversation History	The chatbot maintains session history enabling follow-up questions

FR-12	Responsive UI	All interfaces render correctly on desktop and mobile viewports

3.3 Non-Functional Requirements

The following non-functional requirements were specified at the outset and validated during testing:

- Performance: AI responses must be delivered within 4 seconds under normal network conditions. This was consistently achieved using the Groq API, which returned LLM responses in 1.5–3.5 seconds in testing.
- Scalability: The backend architecture must allow new features (additional event categories, vendor APIs, analytics modules) to be added without restructuring existing endpoints.
- Security: API keys (Groq, MongoDB) must never be exposed in client-side code. All secrets are stored in server-side environment variables (.env files). JWTs must be signed with a strong secret and include an expiry timestamp.
- Usability: Interfaces must be fully responsive and accessible on screens from 320px (small Android devices) to 1440px (desktop). All interactive elements must be reachable without horizontal scrolling.
- Reliability: The system must handle Groq API timeouts gracefully and display clear, user-friendly error messages. MongoDB connection failures must not crash the server.
- Connectivity: The system must function reliably for users on Indian mobile hotspots, where standard MongoDB Atlas SRV DNS lookups may be blocked.

3.4 System Architecture Design

EventIQ uses a classic three-tier web application architecture with an external AI service:

Tier 1 – Presentation Layer: The frontend is a responsive single-page HTML/CSS/JavaScript application. It handles user interaction, renders AI responses, and communicates with the backend exclusively through fetch() API calls to REST endpoints. No business logic or secrets are stored in the frontend.

Tier 2 – Application Layer: The Node.js/Express backend serves as the system's central hub. It exposes REST API endpoints for authentication (register/login), plan management (save/retrieve/delete), and AI chat (proxied to Groq). All sensitive operations — API key usage, database queries, JWT signing — occur in this tier.

Tier 3 – Data Layer: MongoDB Atlas stores two primary collections: users (account credentials and profile information) and plans (saved event plans linked to user IDs). The database is accessed from the backend using the official MongoDB Node.js driver with a direct connection string (mongodb://) to ensure reliability on mobile networks.

External AI Service: The Groq API is called from the backend application layer. The frontend never directly contacts Groq. This architecture ensures that the Groq API key is never exposed in browser source code.

3.5 Database Design

EventIQ uses two MongoDB collections:

Users Collection

Field	Type	Description
_id	ObjectId	Auto-generated unique identifier
name	String	User's full name
email	String	User's email address (unique, indexed)
password	String	bcrypt-hashed password
createdAt	Date	Account creation timestamp

Plans Collection

Field	Type	Description
_id	ObjectId	Auto-generated unique identifier
userId	ObjectId	Reference to the owning user (foreign key)
title	String	Short title derived from the user's event description
content	String	Full AI-generated plan text
city	String	City selected by the user
eventType	String	One of six event category values
budget	Number	User-specified budget in ₹
guestCount	Number	Number of guests
createdAt	Date	Plan creation timestamp

3.6 API Design

The EventIQ backend exposes a RESTful API. Key endpoints are documented below:

Method	Endpoint	Auth Required	Description
POST	/api/auth/register	No	Create a new user account
POST	/api/auth/login	No	Authenticate and receive JWT

GET	/api/plans	Yes (JWT)	Retrieve all plans for the logged-in user
POST	/api/plans	Yes (JWT)	Save a new AI-generated plan
DELETE	/api/plans/:id	Yes (JWT)	Delete a specific plan by ID
POST	/api/chat	Yes (JWT)	Send a message to the AI chatbot (proxied to Groq)

Chapter 4: Methodology

4.1 AI Event Planning Module

At the heart of EventIQ is a conversational AI chatbot powered by the Groq API, which provides access to large language models — specifically LLaMA 3 70B — with extremely low latency inference. A carefully crafted system prompt instructs the model to act as an experienced Indian event planner with deep knowledge of local vendor pricing across major Indian cities.

The system prompt is the most critical design element of the AI module. It establishes the model's persona and constraints:

- The model must identify as an Indian event planning expert familiar with pricing in Bengaluru, Hyderabad, Chennai, Mumbai, Delhi, Kolkata, Pune, and other major cities.
- Responses must include itemised cost breakdowns in Indian Rupees (₹), not in generic global ranges.
- Plans must be tailored to the specific event type — weddings, corporate events, birthday parties, college fests, baby showers, and house parties each have distinct vendor categories.
- The model must acknowledge guest counts and budget constraints explicitly, and flag if the requested budget appears insufficient for the event type.
- Vendor suggestions must be city-specific and realistic.

The module maintains a full conversation history in the browser session so that users can ask follow-up questions like "Can we reduce the catering cost to ₹500 per head?" and receive updated, context-aware responses. Each message in the conversation history is sent to the Groq API as part of the messages array, allowing the model to reference prior context.

On the backend, the /api/chat endpoint receives the conversation history from the frontend, prepends the system prompt, and forwards the complete messages array to the Groq API. The streamed response is returned to the frontend, where it is rendered in the chat interface.

4.2 User Authentication System

EventIQ implements a stateless JWT-based authentication system. The authentication flow operates as follows:

9. Registration: The user submits their name, email, and password to the /api/auth/register endpoint. The backend validates the input, checks for duplicate email addresses, hashes the password using bcrypt (10 salt rounds), and stores the user document in MongoDB.

10. Login: The user submits email and password to /api/auth/login. The backend retrieves the user document by email, compares the submitted password against the stored bcrypt hash, and if valid, signs a JWT containing the user's _id and email with a 24-hour expiry.
11. Token Storage: The JWT is returned to the frontend and stored in localStorage. All subsequent requests to protected endpoints include the token in the Authorization header as a Bearer token.
12. Middleware Verification: A custom Express middleware function extracts and verifies the JWT on every request to protected routes. If the token is missing, malformed, or expired, the middleware returns a 401 Unauthorized response and the request is rejected.
13. Logout: The frontend clears the JWT from localStorage. Since the system is stateless, no server-side session invalidation is required.

Password security is maintained through bcrypt hashing, ensuring that even if the database were compromised, plaintext passwords would not be recoverable. JWT signing uses a strong secret stored in the server's .env file, never hardcoded in source code.

4.3 Saved Plans Module

The Saved Plans module allows authenticated users to persist AI-generated event plans to their MongoDB account for later retrieval. The module operates as follows:

When a user is satisfied with an AI-generated plan, they click the Save Plan button in the chat interface. The frontend sends a POST request to /api/plans with the plan content, title (derived from the first message in the conversation), city, event type, guest count, and budget. The backend middleware verifies the JWT, extracts the userId, and inserts a new document into the plans collection.

The My Saved Plans dashboard retrieves all plans associated with the logged-in user via a GET /api/plans request. The backend queries the plans collection with a { userId } filter and returns the matching documents sorted by creation date (newest first). The frontend renders each plan as a card displaying the plan title, creation date, a truncated preview of the plan content, and action buttons for PDF export and deletion.

Plan deletion is handled by a DELETE /api/plans/:id request. The backend verifies that the plan's userId matches the authenticated user's _id before performing the deletion, preventing users from deleting plans that do not belong to them.

4.4 PDF Export Module

The PDF Export module converts saved event plans into downloadable PDF documents. This feature addresses a clear user need: sharing event plans with family members, vendors, or colleagues who may not have an EventIQ account.

PDF generation is implemented using the jsPDF library on the frontend. When a user clicks the PDF button on a saved plan card, the frontend retrieves the full plan content and passes it to jsPDF, which renders the text with appropriate formatting — including the EventIQ header, plan title, creation date, and full plan content — into a PDF document. The generated PDF is then downloaded directly to the user's device without requiring a server round-trip.

The PDF module handles long plan content through automatic page breaking. When the content exceeds the height of a single page, jsPDF inserts a page break and continues rendering on a new page. Font sizes, line heights, and margins are calibrated to produce professional-looking documents that are clearly readable when printed.

4.5 Budget Editor Module

The Budget Editor allows users to modify the cost line items in a saved plan and see revised totals in real time. This feature is particularly useful when users want to explore trade-offs — for example, reducing the catering budget to increase the decoration allocation — without generating a completely new AI plan.

The editor parses the AI-generated plan text to identify cost line items (structured as item name: ₹amount patterns). These items are rendered as editable input fields in a table layout. As users modify values, a running total is recalculated and displayed. Users can save the modified plan back to their account with the updated figures, or download it as a PDF.

4.6 City-Specific Vendor Suggestions

One of EventIQ's key differentiators from generic AI chatbots is its city-specific vendor suggestion engine. The system is designed to provide vendor recommendations that reflect local market realities rather than generic national averages.

City-specificity is implemented at two levels. At the AI prompt level, the system prompt instructs the Groq model to reference pricing appropriate to the user's specified city. Vendor categories, price ranges, and typical inclusions vary significantly across Indian cities — a wedding venue in Bengaluru commands very different rates than an equivalent venue in a Tier-2 city. The system prompt contains city-specific calibration instructions that guide the model's responses accordingly.

At the application level, a curated vendor category database maps event types to relevant vendor categories for each supported city. When a plan is generated, the frontend displays a contextual vendor suggestion panel populated from this database. This panel provides users with category-level guidance (e.g., "Caterers in Hyderabad typically charge ₹600–₹1,200 per plate for North Indian vegetarian menus") alongside the AI-generated plan.

The cities currently supported by the vendor suggestion engine include Bengaluru, Hyderabad, Chennai, Mumbai, Delhi, Kolkata, Pune, Jaipur, Ahmedabad, and Kochi. Each city's vendor database is structured around the six event categories supported by the platform.

Chapter 5: Module Description

5.1 Frontend Module

The EventIQ frontend is a responsive single-page application built with HTML5, CSS3, and vanilla JavaScript (ES6+). The decision to use vanilla JavaScript rather than a framework like React or Vue was deliberate: it reduces bundle size, eliminates the need for a build pipeline, and makes the application easier to deploy to any static hosting service.

The frontend is organised around four main views, navigated through client-side routing (toggling CSS visibility classes without page reloads):

- Landing Page: Introduces EventIQ with a hero section, an event category grid ("What Can I Plan?"), and a call-to-action button directing users to the chat interface.
- Chat Interface: The primary interaction surface. Contains the AI chat window, event type selector, city selector, guest count and budget inputs, and the chat input field. Conversation history is maintained in a JavaScript array in memory.
- My Saved Plans: A card-based dashboard displaying all plans saved to the user's account. Each card shows the plan title, creation date, a content preview, and PDF/Delete action buttons.
- Authentication Views: Login and registration forms with client-side validation.

The UI design follows principles of clarity and accessibility: white backgrounds, generous whitespace, clear typographic hierarchy, and a consistent blue-and-white colour palette. All interactive elements meet minimum touch target size requirements (48×48px) for mobile usability. CSS custom properties (variables) ensure visual consistency across all views.

5.2 Backend API Module

The EventIQ backend is a Node.js application using the Express.js framework. It is structured around the Model-Controller pattern, with route handlers organised by domain (auth, plans, chat) and shared middleware functions handling cross-cutting concerns like authentication verification and CORS.

The entry point (server.js or index.js) initialises the Express application, configures middleware (CORS, body-parser, helmet for security headers), establishes the MongoDB connection, registers route handlers, and starts the HTTP server on the configured port.

Route files define the API endpoints for each domain:

- routes/auth.js: POST /register and POST /login endpoints.

- routes/plans.js: GET /, POST /, and DELETE /:id endpoints for plan management, all protected by the JWT middleware.
- routes/chat.js: POST / endpoint that proxies chat messages to the Groq API.

The JWT middleware function (middleware/auth.js) is applied to all protected routes. It extracts the token from the Authorization header, verifies it using the JWT secret, and attaches the decoded user payload to the request object for downstream use.

Environment variables — including the MongoDB connection string, Groq API key, and JWT secret — are loaded from a .env file using the dotenv package. This ensures that no secrets appear in source code or version control.

5.3 AI Integration Module (Groq API)

The AI integration module is responsible for communicating with the Groq API to generate event planning responses. Groq provides OpenAI-compatible REST API endpoints, meaning the integration uses the same request format as the widely documented OpenAI Chat Completions API, with the base URL and API key changed to point at Groq's servers.

Each request to the /api/chat endpoint constructs a messages array comprising:

14. The system message: A detailed prompt instructing the model to act as an experienced Indian event planner, specifying the required output format (itemised cost table, vendor suggestions, timeline recommendations), the requirement to use Indian Rupees, and the city-specific calibration instructions.
15. The conversation history: All prior user and assistant messages from the current session, enabling contextually aware follow-up responses.
16. The new user message: The latest input from the user.

The request is sent to the Groq API with the model identifier (e.g., llama3-70b-8192 or mixtral-8x7b-32768) and a maximum token limit. The response is extracted from the API response body and returned to the frontend as a JSON object containing the assistant's message text.

Error handling in this module covers: Groq API authentication failures (invalid API key), rate limiting responses (429 status codes), network timeouts, and malformed response bodies. All errors produce descriptive messages that are logged on the server and returned to the frontend in a user-friendly format.

5.4 Database Module (MongoDB Atlas)

The database module manages the connection to MongoDB Atlas and provides helper functions for all database operations used by the plans and auth modules.

A key architectural decision in this module was the choice to use a direct connection string (mongodb://username:password@host1:port1,host2:port2,host3:port3/database?replicaSet=...) rather than the standard MongoDB Atlas SRV-based connection string (mongodb+srv://...). This decision was made after discovering that DNS SRV lookups — required to resolve the SRV-format connection string — are blocked by Jio and Airtel mobile network DNS resolvers in India. Users accessing the application on mobile hotspots would encounter connection failures when using the SRV format.

By switching to a direct connection string with explicit shard hostnames and port numbers, the application bypasses the SRV DNS lookup entirely and establishes a direct TCP connection to the MongoDB Atlas cluster. This change made the platform reliably accessible on Indian mobile networks with no other changes required.

The database connection is initialised once at application startup and reused for all subsequent requests. Connection pooling is managed by the MongoDB Node.js driver automatically. The module exports the database client object, which is imported by the auth and plans route handlers.

5.5 Authentication Module (JWT)

The authentication module implements user registration, login, and JWT verification. Password hashing uses bcrypt with a work factor of 10, which provides a good balance between security and hashing speed. The bcrypt hash is stored in the users collection; the original password is never stored.

JWT tokens are signed using the HS256 algorithm with a secret key of at least 32 characters, stored in the .env file. Each token payload contains the user's _id and email, and the token expires after 24 hours. The frontend stores the token in localStorage and includes it in the Authorization header of all subsequent requests.

The JWT verification middleware extracts the token, calls jwt.verify() with the secret key, and — if verification succeeds — attaches the decoded payload to req.user for use by downstream route handlers. If verification fails for any reason (expired token, invalid signature, malformed token), the middleware immediately returns a 401 response.

5.6 PDF Generation Module

The PDF generation module uses the jsPDF JavaScript library, loaded in the browser from a CDN. When the user clicks the PDF button for a saved plan, the module:

17. Creates a new jsPDF document instance with A4 page dimensions.

18. Renders the EventIQ header with the platform name and the plan's creation date.
19. Renders the plan title in a larger font size with a separator line.
20. Renders the full plan content, with automatic line wrapping at the page margin and automatic page breaks when content extends beyond the page height.
21. Triggers the browser's file download mechanism, saving the PDF to the user's device with a filename derived from the plan title.

The module handles special characters in plan content — including the Indian Rupee symbol (₹) — by ensuring the correct Unicode characters are passed to jsPDF's text rendering functions.

Chapter 6: Implementation & Testing

6.1 Development Environment and Tools

EventIQ was developed using Visual Studio Code as the primary code editor, with the following extensions: Prettier (code formatting), ESLint (JavaScript linting), REST Client (API testing), and MongoDB for VS Code (database exploration). Git was used for version control throughout the project.

The development environment comprised:

- Frontend: Static HTML/CSS/JS files served via the VS Code Live Server extension during development. No build tool or bundler was required.
- Backend: Node.js runtime (v18.x LTS) with the following npm packages: express, mongoose (MongoDB driver), dotenv, bcryptjs, jsonwebtoken, cors, and node-fetch (for Groq API calls).
- Database: MongoDB Atlas free tier cluster (M0 Sandbox), accessed via direct connection string.
- AI: Groq API accessed via REST with the official Groq JavaScript SDK (or direct fetch calls to the Groq-compatible OpenAI endpoint).
- Testing: Browser developer tools (Chrome DevTools), Postman for API endpoint testing, and manual user testing sessions.

6.2 System Integration

The integration of the frontend, backend, and external services followed a phased approach:

Phase 1 – Single-File Prototype: The initial prototype was a single HTML file that called the Groq API directly from JavaScript using a hardcoded API key. This allowed the team to validate the AI response quality and system prompt design before building the backend infrastructure. While not suitable for production (due to the exposed API key), it provided rapid feedback on the core value proposition.

Phase 2 – Backend Introduction: The Node.js/Express backend was introduced to proxy Groq API calls, eliminating the client-side API key exposure. The frontend was updated to call the local backend endpoint (/api/chat) instead of Groq directly. CORS middleware was configured to allow requests from the frontend origin.

Phase 3 – Authentication and Database: The MongoDB Atlas connection was established, the users collection was created, and the JWT-based auth endpoints were implemented and tested. The frontend was updated with login and registration forms.

Phase 4 – Plans Module: The plans collection was created, and the save/retrieve/delete endpoints were implemented. The My Saved Plans dashboard was built in the frontend and integrated with the backend.

Phase 5 – PDF and Budget Editor: The jsPDF-based PDF export was implemented on the frontend. The budget editor was built as a modal overlay that parses the plan content and renders editable cost line items.

Phase 6 – City-Specific Vendors and Polish: The vendor suggestion engine was implemented, the UI was refined for mobile responsiveness, and the MongoDB connection string was switched from SRV format to direct format to resolve mobile network connectivity issues.

6.3 Functional Testing

Every module was tested independently for correctness, and then integration-tested as part of the complete system. The following test scenarios were executed:

Authentication Testing

- Register with valid credentials: Verified user creation in MongoDB Atlas.
- Register with duplicate email: Verified 409 Conflict response.
- Login with valid credentials: Verified JWT returned and stored correctly.
- Login with incorrect password: Verified 401 Unauthorized response.
- Access protected endpoint without JWT: Verified 401 response.
- Access protected endpoint with expired JWT: Verified 401 response.

AI Chatbot Testing

The chatbot was tested with over 30 distinct event planning queries spanning all six event categories, five cities, and a range of guest counts (10–500) and budgets (₹20,000–₹50,00,000). Selected test cases:

- "Plan a birthday party for 30 guests in Bengaluru with a budget of ₹40,000" — Verified itemised breakdown with city-appropriate pricing.
- "Plan a 3-day wedding for 300 guests in Hyderabad, budget ₹25,00,000" — Verified multi-day plan with venue, catering, decoration, photography, and music breakdowns.
- "I want a small house party for 15 friends, informal, budget ₹8,000" — Verified that the plan scaled appropriately to the low budget and informal context.
- Follow-up question: "Can we reduce catering to ₹200 per head?" after an initial plan — Verified that the model correctly referenced the prior plan in its revised response.

Plans Module Testing

- Save a plan: Verified document creation in the plans collection with correct userId.
- Retrieve saved plans: Verified that only the logged-in user's plans were returned.
- Delete a plan: Verified deletion from MongoDB and removal from the UI.
- Attempt to delete another user's plan: Verified 403 Forbidden response.

PDF Export Testing

- Export a short plan (1 page): Verified correct PDF layout, font rendering, and ₹ symbol.
- Export a long plan (multiple pages): Verified correct page breaks and content continuation.
- PDF filename: Verified that the filename matches the plan title.

6.4 Cross-Browser and Mobile Testing

EventIQ was tested across the following environments:

Environment	Device/Version	Result
Chrome (Desktop)	v124 on Windows 11	Pass – All features functional
Firefox (Desktop)	v125 on Windows 11	Pass – All features functional
Safari (Desktop)	v17 on macOS Sonoma	Pass – PDF export functional
Edge (Desktop)	v124 on Windows 11	Pass – All features functional
Chrome (Android)	v124 on Android 13	Pass – Responsive layout correct
Safari (iOS)	v17 on iPhone 14	Pass – All interactive elements accessible
Chrome (Android)	Jio Hotspot	Pass after SRV→direct fix

6.5 API Integration Testing

All API endpoints were tested using Postman before frontend integration. This allowed the team to verify server-side logic independently of the frontend and identify issues with request format, response structure, and error handling.

The Groq API integration was specifically tested for: correct message format (system + history + user), model identifier validity, response parsing (extracting the assistant content from the choices array), and error scenarios including invalid API key and exceeding context length.

The MongoDB direct connection string was tested across three network environments: campus Wi-Fi (pass), Jio mobile hotspot (pass after switch from SRV to direct), and Airtel mobile hotspot (pass after switch).

6.6 User Testing Results

Ten users tested EventIQ by planning events across different categories and cities. Results were collected through structured feedback sessions:

Metric	Result	Target
AI cost estimates rated 'accurate' or 'very accurate'	88%	≥ 80%
Users who found the UI 'easy' or 'very easy' to use	90%	≥ 85%
Average AI response time (Groq API)	2.3 seconds	≤ 4 seconds
Plans saved successfully (no data loss)	100%	100%
PDF exports rated 'readable' or 'very readable'	95%	≥ 90%
Users who would recommend EventIQ	85%	≥ 80%

The most commonly requested additional features were: voice input for the chatbot, WhatsApp sharing for PDF plans, and a vendor contact directory. All three are noted for future development.

Chapter 7: Results & Analysis

7.1 Performance Analysis

EventIQ was evaluated across several performance dimensions during the testing phase. All primary performance targets were met or exceeded.

AI Response Latency: The Groq API consistently returned complete event planning responses in 1.5 to 3.5 seconds for typical queries (event plans of 400–800 words). This is significantly faster than GPU-based LLM providers, where comparable queries typically take 5–15 seconds. The low latency is critical for a conversational interface — responses that arrive in under 4 seconds maintain the feeling of a natural conversation.

API Success Rate: Across 150+ API calls made during testing (including chatbot queries, plan saves, and auth operations), the system achieved a 97.3% success rate. The 2.7% failure rate was attributable to Groq API rate limiting during intensive testing sessions — not to application-level bugs. All failures produced appropriate error messages in the UI.

Database Performance: MongoDB read and write operations completed in under 50ms in all test cases. The direct connection string approach introduced no measurable latency overhead compared to the SRV approach.

Frontend Load Time: The application's initial page load (including all CSS and JavaScript) completed in under 1.8 seconds on a 4G mobile connection. Since no framework bundle is required, the total JavaScript payload is well under 100KB.

7.2 Sample Outputs

The following examples illustrate the quality and structure of EventIQ's AI-generated plans:

Example 1: Birthday Party

User input: "Plan a birthday party for 50 guests in Bengaluru with a budget of ₹80,000"

The AI generated a complete plan including: Venue hire (₹15,000 for a banquet hall in Koramangala), catering (₹600/plate for a buffet = ₹30,000), birthday cake (₹3,000 for a 3kg custom cake), decoration (₹8,000 for balloon and floral setup), DJ and music (₹7,000), photography (₹8,000 for 4 hours), invitations and miscellaneous (₹4,000), and contingency (₹5,000). Total: ₹80,000. The plan also included a timeline for the 3-hour event and vendor category suggestions specific to Bengaluru.

Example 2: Wedding

User input: "Plan a wedding for 200 guests in Hyderabad, budget ₹15,00,000"

The AI generated a comprehensive wedding plan spanning pre-wedding, wedding day, and reception. Breakdown included: Venue (₹3,50,000 for a banquet hall), catering (₹700/plate for 200 guests across 2 days = ₹2,80,000), decoration (₹2,00,000), photography and videography (₹1,50,000), bridal makeup and mehendi (₹80,000), DJ and band (₹1,20,000), invitations and stationery (₹40,000), accommodation for family (₹1,00,000), transportation (₹60,000), honeymoon contribution (₹1,00,000), and miscellaneous (₹1,20,000). Total: ₹14,00,000, with ₹1,00,000 contingency.

7.3 Limitations and Challenges

The following limitations and challenges were identified during development and testing:

22. AI Estimate Variance: AI-generated cost estimates are approximations based on the model's training data and the system prompt's calibration. Hyperlocal pricing — such as venue rates in specific neighbourhoods or during peak wedding season — may not be accurately reflected. Users should treat AI estimates as starting points for vendor negotiations, not as final quotes.
23. Mobile Network Connectivity: The standard MongoDB Atlas SRV connection string is blocked on Jio and Airtel DNS resolvers. The solution (direct connection string) is effective but requires manual extraction of shard hostnames from the Atlas console — a non-trivial step that future deployments should automate through a connection string management tool.
24. No Real-Time Vendor Data: The city-specific vendor suggestion engine is based on curated static data, not real-time API calls to vendor platforms. Prices change seasonally and may become stale. Integration with live vendor APIs is planned for a future phase.
25. Single-Session Conversation History: The chatbot's conversation history is stored in browser memory and is lost when the user closes the tab. Users cannot resume a planning conversation from where they left off. Persistent conversation storage is a planned future feature.
26. PDF Formatting Limitations: The jsPDF-based PDF export handles basic text content well but does not render HTML tables or formatted cost breakdowns from the AI response. The PDF output is plain text with line breaks, which is functional but not as visually polished as a purpose-built PDF renderer would produce.

Chapter 8: User Interface

EventIQ's user interface is designed around three principles: clarity, accessibility, and speed. The design was informed by Apple's Human Interface Guidelines and Google's Material Design principles, adapted for the specific context of a conversational event planning application.

8.1 Design Language

The visual design uses a consistent blue-and-white colour palette throughout:

- Primary Blue (#0071e3): Used for primary action buttons, links, and brand accents. This is the same blue used by Apple's website, chosen for its associations with trust and professionalism.
- White (#FFFFFF): Background for all main content areas, creating a clean, uncluttered visual environment.
- Light Grey (#F5F5F7): Background for the overall page and card containers, providing subtle visual hierarchy without harsh contrast.
- Dark Grey (#1D1D1F): Primary text colour, ensuring excellent readability against white and light grey backgrounds.
- Medium Grey (#6E6E73): Secondary text for metadata such as dates and subtitles.

Typography uses the system font stack (-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial), which ensures the best available font on each operating system without requiring a web font download. This reduces page load time and ensures consistent rendering across platforms.

8.2 Landing Page

The landing page introduces EventIQ to new visitors with a hero section featuring the platform name, a one-line value proposition ("AI-powered event planning for India"), and a prominent 'Start Planning' call-to-action button. Below the hero, a six-card grid presents the supported event categories — Weddings, Birthday Parties, Corporate Events, College & Cultural Events, Baby & Bridal Showers, and House Parties — each with a representative emoji, category name, and one-line description.

Clicking any category card takes the user directly to the chat interface with that category pre-selected, reducing the steps required to start planning. This pattern — reducing friction from intent to action — is a core usability principle applied throughout the interface.

8.3 Chat Interface

The chat interface is the primary interaction surface of EventIQ. It is divided into two panels on desktop: a left panel containing the event configuration inputs (city, event type, guest count, budget) and a right panel containing the chat window.

The chat window displays the conversation history in a familiar messaging format: user messages appear right-aligned in blue bubbles, and AI responses appear left-aligned with a white background. A typing indicator (animated ellipsis) appears while the Groq API is processing the request. The input field is pinned to the bottom of the window with an auto-resizing textarea that expands to accommodate multi-line inputs.

A 'Save Plan' button appears in the chat toolbar after the first AI response is received, allowing users to save the current plan to their account. A 'New Chat' button clears the conversation history and resets the interface for a fresh planning session.

8.4 My Saved Plans Dashboard

The My Saved Plans dashboard presents the user's saved plans in a responsive card grid. On desktop, cards are displayed in a three-column grid; on mobile, they reflow to a single column. Each card displays:

- The plan title (derived from the user's first message in the planning session)
- The creation date in a human-readable format (e.g., 20 Apr 2026)
- A truncated preview of the plan content (approximately 100 characters)
- A PDF export button and a Delete button

The currently highlighted card (if any) is visually distinguished with a blue border and subtle shadow elevation, as seen in the application screenshots. Empty state handling — when the user has no saved plans — displays a friendly message and a link to start planning.

8.5 Mobile Responsiveness

The mobile layout reorganises the desktop two-panel chat interface into a single-column layout. The event configuration inputs collapse into a scrollable section above the chat window. The card grid reflows to a single column. All buttons meet minimum touch target size requirements (48×48px).

The application was specifically tested on small-screen Android devices (320px viewport width) to ensure that no horizontal scrolling is required and that all functionality remains accessible.

Chapter 9: Deployment & Execution

9.1 Local Development Setup

To run EventIQ locally, the following steps are required:

27. Install Node.js (v18 LTS or later) from nodejs.org.
28. Clone or download the EventIQ repository to a local directory.
29. In the project root directory, run npm install to install all backend dependencies.
30. Create a .env file in the project root with the following variables: MONGO_URI (the direct MongoDB connection string), JWT_SECRET (a strong random string), GROQ_API_KEY (the Groq API key from console.groq.com), and PORT (the port for the Express server, defaulting to 3000).
31. Run node server.js (or npm start) to launch the backend server.
32. Open the frontend/index.html file in a browser, or configure the Express server to serve static frontend files from the public/ directory.

During development, the team used the nodemon package (npm install -D nodemon) to automatically restart the server when source files changed, significantly accelerating the development iteration cycle.

9.2 Production Deployment Options

Backend Deployment (Node.js/Express)

The EventIQ backend can be deployed to any Node.js-compatible hosting platform. Recommended options for Indian deployments include:

- Railway (railway.app): Automatic Node.js detection, environment variable management, and free tier with sufficient resources for a student project.
- Render (render.com): Similar to Railway, with automatic deployments from GitHub and a generous free tier.
- Heroku: A well-established platform with extensive documentation, suitable for educational projects.
- AWS EC2 or DigitalOcean Droplet: For production deployments requiring greater control over server configuration.

Frontend Deployment

If the frontend is served separately from the backend (as static files), it can be deployed to:

- Netlify: Free CDN hosting with automatic HTTPS and continuous deployment from GitHub.

- Vercel: Similar to Netlify, with excellent performance for static sites.
- GitHub Pages: Free hosting directly from a GitHub repository, suitable for simple static frontends.

In production, the frontend's API base URL must be updated to point to the deployed backend URL rather than localhost.

9.3 Environment Configuration

Production deployments require the following environment variables to be configured on the hosting platform:

Variable	Description
MONGO_URI	Direct MongoDB connection string (mongodb://user:pass@host1:port1,.../?replicaSet=...)
JWT_SECRET	A random string of at least 32 characters used to sign JWT tokens
GROQ_API_KEY	API key from console.groq.com for AI inference
PORT	HTTP port for the Express server (default: 3000; hosting platforms often set this automatically)
NODE_ENV	Set to 'production' to disable debug logging and enable production error handling

9.4 System Requirements

Server Requirements

- Node.js v18 LTS or later
- npm v9 or later
- Minimum 512 MB RAM (1 GB recommended for production)
- Internet access for MongoDB Atlas and Groq API calls

Client Requirements

- Any modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- JavaScript enabled
- Internet connection (4G or better for acceptable AI response times)

Development Environment

- Visual Studio Code with Prettier and ESLint extensions
- MongoDB for VS Code extension for database exploration
- Postman or REST Client for API testing

Chapter 10: Advantages of the System

10.1 Technical Advantages

EventIQ's architecture offers several significant technical advantages:

- Full-Stack Separation of Concerns: The strict separation between frontend, backend, and AI service layers makes the system modular, maintainable, and independently scalable. Changes to the AI provider (e.g., switching from Groq to a different LLM API) require changes only in the chat route handler, not in the database or authentication layers.
- Stateless Authentication: JWT-based authentication eliminates server-side session storage, making the backend horizontally scalable. Multiple backend server instances can handle requests from the same user without requiring session affinity or a shared session store.
- Mobile Network Resilience: The direct MongoDB connection string approach solves a real deployment problem specific to the Indian context. This solution is not widely documented and represents a genuine contribution to practical deployment knowledge for Indian developers.
- AI Provider Agnosticism: Because the Groq API uses the OpenAI-compatible REST format, switching to OpenAI, Anthropic's Claude API, or any other OpenAI-compatible provider requires only a URL and authentication key change, with no changes to the application logic.
- Zero-Build Frontend: The vanilla JavaScript frontend requires no build pipeline, Webpack configuration, or Node modules on the client side. This dramatically simplifies deployment and makes the frontend trivially easy to host anywhere.

10.2 Practical Benefits

- Indian Market Focus: Unlike generic AI chat interfaces, EventIQ's AI prompting is specifically calibrated for Indian cities, event types, pricing norms, and vendor categories. This produces materially more useful outputs for Indian users.
- Plan Persistence: The ability to save, retrieve, and revisit event plans across sessions transforms EventIQ from a one-shot AI tool into a genuine planning companion. Users can return to their plans days or weeks later to review, compare, or update them.
- PDF Export for Sharing: The PDF export feature bridges the gap between digital planning and real-world execution. Users can share professional-looking plan documents with family members, vendors, or colleagues who do not use the platform.
- Budget Editor: The budget editor allows users to explore trade-offs within their plans without regenerating a complete AI response. This reduces API usage and gives users direct control over their event budget allocations.

- Rapid Deployment: The combination of Railway/Render backend hosting, MongoDB Atlas database, and Netlify/Vercel frontend hosting means a complete production deployment can be achieved within a few hours at zero cost.

Chapter 11: Limitations of the System

11.1 Technical Limitations

- Static Vendor Database: The city-specific vendor suggestion engine is based on curated static data rather than live vendor APIs. This means vendor information may become outdated as market prices change. Real-time integration with vendor platforms like WedMeGood, Sulekha, or JustDial is required for production-grade vendor recommendations.
- No Conversation Persistence: Conversation history is stored in browser memory and lost when the user closes or refreshes the tab. Users cannot resume a planning conversation from a prior session. Implementing persistent conversation storage in MongoDB would resolve this limitation.
- jsPDF Rendering Limitations: The client-side PDF generation using jsPDF produces plaintext output. It does not render HTML tables, bullet lists, or formatted cost breakdowns from the AI response in their original visual structure. A server-side PDF generation solution (using Puppeteer or a headless browser) would produce significantly better-formatted output.
- No Webhook or Real-Time Features: The current architecture uses a request-response model for all operations. There is no WebSocket or Server-Sent Events implementation for real-time features such as live AI response streaming.
- No Image or Media Support: The platform does not allow users to upload inspiration images, venue photos, or decoration references that could be incorporated into AI planning prompts.

11.2 Practical Constraints

- AI Hallucination Risk: Like all large language models, the Groq-hosted models can occasionally generate inaccurate pricing or vendor information. Users should verify all estimates with actual vendors before committing to a budget.
- API Cost at Scale: The Groq API is free within generous rate limits but incurs costs at higher usage volumes. A production platform with thousands of daily active users would require a paid API plan and usage monitoring.
- Limited to Six Event Categories: The current platform supports six event categories. Requests for categories not in this list (e.g., funerals, religious ceremonies, sports events) may produce less accurate AI responses because the system prompt is optimised for the six defined categories.

Chapter 12: Future Enhancements

12.1 Near-Term Enhancements (0–6 months)

The following enhancements are planned as immediate next steps following the current project phase:

- Persistent Conversation History: Store conversation sessions in MongoDB, allowing users to resume planning conversations from any device. Each session would be linked to the user's account and associated with a specific event plan.
- Streaming AI Responses: Implement Server-Sent Events (SSE) or WebSockets to stream AI responses token-by-token as they are generated, rather than waiting for the complete response before displaying it. This would eliminate the perceived waiting time and make the chatbot feel more responsive.
- Improved PDF Generation: Replace client-side jsPDF with a server-side Puppeteer-based PDF generator that renders the full HTML of the event plan — including formatted tables, bullet points, and colour coding — into a professional PDF document.
- Plan Comparison: Allow users to generate multiple plans for the same event (e.g., a premium plan vs. a budget plan) and view them side-by-side in a comparison table.
- WhatsApp Sharing: Add a "Share on WhatsApp" button that generates a formatted text summary of the event plan and opens a WhatsApp chat with the content pre-filled, enabling easy sharing with family and vendors.

12.2 Medium-Term Enhancements (6–18 months)

- Mobile Application: A React Native application for iOS and Android would bring EventIQ's features to a dedicated mobile experience, with additional capabilities such as push notifications for plan reminders, camera integration for venue photo uploads, and offline access to saved plans.
- Live Vendor Integration: API integration with Indian vendor discovery platforms (WedMeGood, Sulekha, JustDial) to provide real-time vendor listings, contact details, and pricing for the user's city and event type. This would transform EventIQ's vendor suggestions from static guidance to actionable vendor discovery.
- Payment Integration: Integration with Razorpay or PayU to enable users to make advance payments to vendors directly through the EventIQ platform, creating a marketplace model with potential commission revenue.
- Multi-Language Support: Tamil, Telugu, Hindi, Kannada, and Marathi interface translations would significantly expand EventIQ's addressable market across India. AI plan generation in regional languages would further personalise the experience for non-English speakers.

- Collaborative Planning: Allow multiple users (e.g., the bride's family and the groom's family) to collaborate on the same event plan simultaneously, with real-time updates and a shared comment thread.

12.3 Long-Term Vision (18+ months)

- Vendor Marketplace: A full-featured vendor marketplace where vendors can list their services, showcase portfolios, receive booking requests, and manage their schedules. EventIQ would serve as both the consumer planning interface and the vendor management platform.
- AI Voice Interface: Integration of speech-to-text (Web Speech API or a dedicated ASR service) and text-to-speech to enable voice-based event planning. Users could describe their event requirements verbally and receive spoken plan summaries.
- Predictive Budget Analytics: A machine learning model trained on historical event plan data to predict budget sufficiency, identify cost overrun risks, and suggest optimisation strategies before the user finalises their plan.
- Event Day Management: A day-of event management module with a real-time timeline, vendor check-in tracking, issue logging, and guest management features to support event organisers throughout the execution phase.

Chapter 13: Problem Statement

13.1 Detailed Problem Definition

India's event planning industry is characterised by fragmentation, information asymmetry, and a near-complete absence of intelligent digital tools accessible to everyday consumers. The problems are structural and deeply embedded in how events are planned across the country:

First, cost opacity is pervasive. A family planning a wedding in Bengaluru has no reliable way to know what a reasonable catering cost per plate looks like, how much a mandap decorator typically charges, or what range of quotes to expect from a wedding photographer. This information asymmetry systematically disadvantages consumers and enables overcharging by vendors who understand that most clients are comparing only the few quotes they have personally solicited.

Second, planning is fragmented across tools and channels. A wedding is planned across WhatsApp groups (for family coordination), Excel spreadsheets (for budget tracking), Google search (for vendor discovery), and verbal communication with vendors — with no single tool integrating all of these activities. The cognitive load of managing this fragmentation is significant, particularly for first-time event planners.

Third, existing digital tools are not localised for India. Most AI-powered planning tools are designed for Western markets and produce cost estimates in USD based on North American or European vendor pricing. These tools are not useful for an Indian user planning an event in Hyderabad or Chennai, where market structures, vendor categories (pandit, mehendi artist, nadaswaram performer), and price ranges are fundamentally different.

Fourth, the technical infrastructure has not been adapted for India's connectivity landscape. Standard deployment architectures for modern web applications assume reliable broadband connectivity and functional DNS resolution for SRV records. These assumptions do not hold for the hundreds of millions of Indians who primarily access the internet through Jio or Airtel mobile data, where SRV-based DNS lookups for MongoDB Atlas are blocked at the network level.

EventIQ addresses all four of these structural problems. It provides cost transparency through AI-generated itemised plans. It integrates planning into a single platform. It is specifically calibrated for Indian cities and event types. And it is architecturally designed to work reliably on Indian mobile networks.

13.2 Challenges Identified and Resolved

The following specific technical challenges were identified and resolved during development:

Challenge 1: MongoDB Atlas SRV DNS Failure on Mobile Networks

Symptom: The application failed to connect to MongoDB Atlas when accessed on Jio and Airtel mobile hotspots, producing a DNS resolution error for the _mongodb._tcp SRV record.

Root Cause: Indian mobile network DNS resolvers do not forward SRV record lookups, blocking the mongodb+srv:// connection string resolution.

Resolution: Extracted the individual shard hostnames and port numbers from the MongoDB Atlas connection string generator (available in the Atlas console under "Connect") and constructed a direct mongodb:// connection string with explicit hostnames. This bypasses SRV DNS entirely and establishes a direct TCP connection.

Challenge 2: API Key Security

Symptom: The initial single-file prototype included the Groq API key directly in client-side JavaScript, exposing it to any user who inspected the page source.

Root Cause: No backend layer existed to proxy API calls.

Resolution: Introduced the Node.js/Express backend as an API proxy. The Groq API key is now stored in the server's .env file and never transmitted to the client.

Challenge 3: Frontend/Backend Inconsistency

Symptom: During one development iteration using an agent-based code generation tool, the generated frontend and backend code were inconsistent — API endpoint paths and request formats did not match.

Root Cause: The code generation tool switched LLM models mid-task, resulting in the frontend and backend being generated with different assumptions about the API contract.

Resolution: The frontend was replaced entirely with a manually written implementation that matched the backend API contract. This experience reinforced the importance of defining the API contract (endpoint paths, request/response formats) explicitly before generating or writing code for either tier.

Conclusion

This project set out to demonstrate that artificial intelligence can meaningfully improve the way events are planned, budgeted, and experienced across India — for everyday consumers and professional organisers alike. We are confident it has done that.

EventIQ was successfully developed as a full-stack AI-powered event planning platform. The conversational AI chatbot delivers accurate, context-aware event cost estimates and structured plans in Indian Rupees, calibrated to city-specific pricing across major Indian cities. The user account system allows plans to be saved, retrieved, and revisited across sessions. The PDF export feature bridges digital planning and real-world execution. And the budget editor gives users direct control over their event allocations.

Beyond its features, the project delivered several technical contributions of broader relevance to the Indian developer community. The identification and resolution of the MongoDB Atlas SRV DNS failure on Indian mobile networks — and the direct connection string workaround — is a practical solution to a real problem that is not widely documented. The JWT authentication implementation, the Groq API integration pattern, and the frontend/backend API proxy architecture together form a reusable template for full-stack AI application development.

The testing phase produced encouraging results: 88% of AI-generated cost estimates were rated accurate by users, 90% of users found the interface easy to use, and average AI response times of 2.3 seconds were achieved — well within the 4-second threshold for conversational interfaces.

EventIQ is not a finished product — it is a foundation. The platform is designed to grow: with live vendor APIs, payment integration, a mobile application, multi-language support, and a vendor marketplace all identified as high-priority next steps. The architecture supports each of these extensions without requiring structural changes to the existing codebase.

We believe EventIQ demonstrates something important: that AI can make professional-grade event planning accessible to everyone in India, not just those with big budgets or industry connections. The tools exist. The technology works. The market is enormous. EventIQ is our contribution to closing that gap.

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